

Flood Early Warning from Low-Cost River Sensors

by

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Abstract

Flash floods on small rivers give villages little time to move people and livestock. National gauges are sparse and report every three hours. This thesis studies whether a network of low-cost ultrasonic sensors, read every five minutes, can give a useful warning for a single river basin. A gradient boosting model trained on two monsoon seasons of readings predicts the water level six hours ahead with a mean error of 9.4 cm.

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Chapter 1

Introduction

1.1 Motivation

In the north-eastern basin, water can rise two metres in a night. Families who live on the banks rely on shouted warnings from upstream neighbours. The national gauge network has one station for the whole river, and its readings reach the district office hours after they are taken. A denser network of inexpensive sensors could close this gap.

Commercial river gauges cost several thousand dollars each and need a technician to install them. In the last decade ultrasonic range finders, small solar panels and mobile modems have become cheap enough that a complete measuring unit can be built for less than a hundred dollars. Units of this kind are less accurate than a commercial gauge, but a district can afford many of them, and many imperfect readings may be worth more than one good reading that arrives late.

A warning is only useful when it reaches people in time to act. Interviews carried out in four villages before this work began suggested that families need about four hours to move cattle and stored rice to higher ground. A forecast that looks six hours ahead would therefore leave a margin of two hours for the message to travel.

The river studied here is 94 km long and falls 310 m between the hills and the plain. Its upper reaches respond to heavy rain within two hours, while the lower reaches, where most people live, respond more slowly. This delay between upstream and downstream levels is what makes a forecast from upstream sensors possible at all.

1.2 Research Questions

This thesis asks how accurately the water level at a village can be predicted six hours ahead from upstream sensor readings, and how many sensors are needed before additional ones stop improving the forecast.

A third question concerns reliability. Inexpensive sensors fail, lose power and are occasionally stolen. The thesis therefore also asks how much the forecast degrades when some of the units are silent, since a warning system that only works when every unit is healthy would be of little use in the monsoon.

Chapter 2

Literature Review

Physically based hydrological models describe a basin with equations for rainfall, runoff and channel flow (Beven, 2012). They need detailed terrain and soil data that are rarely available for small rivers. Data-driven models learn the relation between upstream and downstream levels directly from observations (Mosavi et al., 2018). Most published studies use government gauges; few evaluate sensors that cost less than a hundred dollars each.

Chapter 3

Methodology

3.1 Sensor Network

Fourteen ultrasonic sensors were mounted on bridges along 62 km of the river. Each unit measures the distance to the water surface every five minutes and sends the reading over the mobile network. Readings were collected from May 2022 to October 2023.

Figure 3.1: Location of the fourteen sensors along the river.

3.2 Forecasting Model

A gradient boosting regressor takes the last twelve hours of readings from every upstream sensor and predicts the level at the target village six hours later. The first season is used for training and the second for testing.

Chapter 4

Results

On the second season the model predicts the six-hour level with a mean absolute error of 9.4 cm, compared with 31.7 cm for a persistence baseline. Removing sensors one at a time shows that eight sensors give almost the same error as fourteen. All eleven events in which the river crossed the danger mark were predicted at least four hours in advance.

Table 4.1: Mean absolute error by forecast horizon.

Chapter 5

Conclusion

This chapter summarises what the thesis found, states its limitations and suggests directions for further study.

5.1 Summary of Findings

A network of fourteen low-cost sensors, read every five minutes, is enough to predict the water level at a downstream village six hours ahead with an error below ten centimetres. Eight well-placed sensors perform almost as well as fourteen, which lowers the cost of a warning system for a small river to a level that a district council can afford.

The forecast remained usable when up to three units were silent, provided that the two units closest to the hills kept working. This points to a practical rule for anyone who installs such a network: spend the maintenance budget on the uppermost sensors first.

All eleven occasions on which the river crossed the danger mark in the second season were predicted at least four hours in advance, and the model raised two false alarms. Villagers who were shown the forecasts said that two false alarms in a season would not stop them from acting on a warning.

Taken together, the results answer the three research questions. The level can be predicted six hours ahead to within ten centimetres, eight sensors are enough, and the system tolerates the loss of a few units.

5.2 Limitations

The findings rest on two monsoon seasons from one river basin. Neither season contained a flood as severe as that of 2017, so the behaviour of the model in extreme events is untested.

Sensor outages removed 6% of the readings, and the gaps were filled by interpolation, which may hide short surges. The results therefore should not be taken as evidence of generalization to rivers with dams or tidal influence.

5.3 Future Work

Further research should test the approach on at least two other basins and through a season with a severe flood. Rainfall forecasts from satellite products could be added as inputs to extend the warning time beyond six hours. A field trial with the district disaster committee would show whether the warnings change what people actually do.

References

- Beven, K. (2012). *Rainfall-Runoff Modelling: The Primer* (2nd ed.). Wiley-Blackwell.
- Mosavi, A., Ozturk, P., and Chau, K. (2018). Flood prediction using machine learning models: Literature review. *Water*, 10(11), 1536.
- Kratzert, F., Klotz, D., Brenner, C., Schulz, K., and Herrnegger, M. (2018). Rainfall-runoff modelling using long short-term memory networks. *Hydrology and Earth System Sciences*, 22, 6005-6022.
Code: https://github.com/kratzert/ealstm_regional_modeling

Appendix A

Sensor Wiring

Each unit consists of an ultrasonic range finder, a microcontroller with a mobile modem and a solar panel with a battery. The design files are kept at <https://osf.io/q7xk2> for anyone who wishes to build the sensors.